

Vinayak Gajjewar

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Research statement I am interested in the intersection of IoT, distributed systems, and scalable spatial analytics. Specifically, I want to bring geospatial and spatio-temporal applications to resource-constrained computing environments in a scalable and fault-tolerant manner.

Education *Ph.D., Computer Science* Expected June 2028
University of California, Santa Barbara

- GPA: 4.0/4.0
- Advisors: Chandra Krintz and Rich Wolski
- Research topics: Internet of Things, distributed systems, geospatial computing
- Selected coursework: Scalable Internet Services (291A), Runtime Systems (263), Blockchain and Distributed Systems (293B), Operating Systems (270), Program Analysis (260)

Bachelor of Science, Computer Science Graduated June 2023
University of California, Riverside

Adrian Wilcox High School (*Santa Clara, CA*) June 2019

Skills Unix/Linux, Python, Django, C, C++, Docker, Git, Java, Scala, Apache Spark, Hadoop, Node.js, JavaScript, TypeScript, MongoDB, SQL, AWS Lambda

Publications Gajjewar, Vinayak, Rich Wolski, and Chandra Krintz. "Ariadne: A Resilient IoT Storage Service for Geospatial Computing." **17% acceptance rate!**

Gajjewar, Vinayak, Rich Wolski, and Chandra Krintz. "RIoTstore: Resilient Data Storage for Spatial IoT Applications." 2025 IEEE Annual Congress on Artificial Intelligence of Things (AIoT). IEEE, 2025.

Singla, Samriddhi, Ayan Mukhopadhyay, Michael Wilbur, Tina Diao, Vinayak Gajjewar, Ahmed Eldawy, Mykel Kochenderfer, Ross Shachter, and Abhishek Dubey. "Wildfiredb: An open-source dataset connecting wildfire occurrence with relevant determinants." In NeurIPS Thirty-fifth Annual Conference on Neural Information Processing Systems. 2021.

Experience *Software Products Intern* Summer 2024
Esri Inc., Redlands, CA

- Used TypeScript, CloudFormation, and Electron to develop an application for deploying ArcGIS Enterprise to AWS.

Undergraduate Researcher Summer 2023
UC Riverside Big-Data Lab, Riverside, CA

- Contributed to the development of Raptor, a Raster + Vector query processing engine written in Java and Spark for manipulating and visualizing geospatial data.

	<i>Software Products Intern</i> Summer 2022 Esri Inc., Redlands, CA • Used Node.js to write a framework for connecting remote data sources (e.g., databases, APIs) to the Esri software ecosystem. • Wrote technical documentation and code samples for new features of ArcGIS Enterprise.
	<i>Research Fellow</i> May 2021 - May 2022 Digital Agriculture Fellowship, Riverside, CA • Used Maven and Apache Spark to build a scalable analytics system that uses satellite data to compute wildfire spread, resulting in a 2 order of magnitude performance increase over the state of the art.
	<i>Instructor</i> Summer 2020 iD Tech Camps, Santa Clara, CA • Tutored 50+ K-5 children on various topics in computer science, from basic programming concepts to video game development.
	<i>Computer Science Intern</i> Summer 2018 SchoolCity Inc., Santa Clara, CA • Developed a data analytics application that extrapolates patterns in School City product usage across school districts using MongoDB and Express.js.
Teaching experience	<i>UCSB CMPSC 170: Operating Systems</i> Winter 2024, Spring 2024 Teaching Assistant • Worked with Profs. Rich Wolski (W24) and Tao Yang (S24) • Subjects: processes, inter-process communication, I/O, file systems, memory management • Debugged OS implementations, held office hours and remedial sessions, <i>UCSB CMPSC 190B: IoT Systems</i> Fall 2023 Teaching Assistant • Worked with Prof. Chandra Krintz • Subjects: IoT fundamentals, software architectures, communication protocols, security concerns, distributed + multi-tier (sensors+edge+cloud) programming • Managed hardware (Raspberry Pi + Arduino), debugged implementations, held office hours and remedial sessions
Presentations	2025 — Spatial awareness for scalable IoT systems. (2025). Major Area Exam. Santa Barbara, CA. 2023 — Implementing a Distributed Evapotranspiration Model. (2023). 2023 UCR Undergraduate Research & Creative Activities Symposium. Riverside, CA. 2021 — Increasing the Efficiency of Geospatial Data Processing. (2021). Research in Science & Engineering Symposium. Riverside, CA.
Awards & honors	2023 — ASA, CSSA, SSSA Outstanding Senior (19 seniors recognized nationally) 2021 — Digital Agriculture Fellowship (Artificial Intelligence for Sustainable Agriculture)
Selected press	Ober, Holly. (December 8, 2021.) Wildfire dataset could help firefighters save lives and property. UCR News Archive. https://news.ucr.edu/articles/2021/12/08/wildfire-dataset-could-help-firefighters-save-lives-and-property